33rd Tree Labs: The Singularity Protocol

A Byzantine-Resilient Verifiable DePIN Compute Fabric on Solana

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ABSTRACT

As autonomous artificial intelligence agents assume direct fiduciary stewardship over capital, centralized cloud infrastructure introduces fatal vulnerabilities: memory bus snooping, unverified output manipulation, and administrative censorship. While proof-of-work pioneered trustless coordination, evaluating arbitrary hash preimages squanders gigawatts of power on zero computational utility. We present 33rd Tree Labs and the Singularity Protocol, an off-chain verifiable compute fabric settled natively on Solana. The network replaces hash lotteries with Proof of Intelligence (PoI), an unforgeable physical GPU admission gate requiring bit-exact evaluations of standard transformer layers verified via BN254 polynomial commitments, while slot proposals scale with stake among verified nodes. To eliminate hardware vendor single points of failure, confidential execution is governed by a Heterogeneous Multi-TEE Attestation Quorum (Intel TDX, AMD SEV-SNP, and NVIDIA CC) paired with an autonomous Succinct ZK-Fallback engine. Model weights and activations execute under Calibrated SmoothQuant FP8 (E4M3/E5M2) bit-exact integer reduction trees, guaranteeing bounded error $\Delta\mathcal{L} \leq 1.2 \times 10^{-4}$ on 70B+ parameter Mixture-of-Experts (MoE) topologies. Validator overlays embed into a non-Euclidean Exponential Plunge Hyperboloid manifold $\mathbb{H}^2$ that strictly bounds network diameter to ≤ 8 hops under 40% adversarial churn and exponentially raises Sybil Eclipse attack costs. Client inference latency is decoupled through a Three-Tier Finality Pipeline delivering instant speculative execution (≤ 450 ms) backed by on-chain slashing insurance, 12s BFT consensus finality, and 48-hour dispute bisection. Staking collateral, slashing, and commercial fee distributions are anchored to Solana via the native $33RD token (hard-capped at 33,333,333 units), incorporating dynamic difficulty retargeting with Moore's Law hardware yield decay and an automated 10% buyback-and-burn engine routed to the Solana Incinerator.

1. Introduction

Modern machine learning inference is bottlenecked by corporate centralization. Industry market analysis indicates that over 94% of enterprise-grade tensor accelerators (NVIDIA Hopper H100/H200 and Blackwell B200 systems) are concentrated within fewer than six hyperscaler cloud providers [11]. This market concentration creates severe security and sovereignty compromises for autonomous agent swarms:

1. **Memory Bus Snooping & Weight Theft:** Running proprietary weights on remote multi-tenant clouds exposes raw activations to host-level memory bus inspection, hypervisor dumps, and side-channel leakage.
2. **Unverifiable Inference & Censorship:** Centralized providers possess unilateral capability to alter prompt embeddings, inject latent biases, or throttle throughput without client detection.
3. **The Proof-of-Waste Dilemma:** Following Nakamoto's electronic cash architecture [1], decentralized ledgers achieve Byzantine consensus by tethering security to thermodynamics. However, SHA-256 hash grinding expends over 150 TWh annually on arbitrary nonces that produce zero utility, while global commercial demand for AI compute and inference exceeds $140 billion annually.

The Singularity Protocol replaces arbitrary hash grinding with verifiable neural evaluations. Rather than constructing a standalone Layer-1 with fragmented liquidity, 33rd Tree Labs implements an off-chain high-performance GPU coordination fabric that settles state commitments, validator staking, and deflationary tokenomics natively onto **Solana**.

1.1 Core Mathematical Notation

Throughout this specification, we employ standardized cryptographic and tensor notation: $\mathbb{F}_p$ denotes the scalar field of the BN254 elliptic curve ($p \approx 2^{254}$); $\mathbb{G}_1, \mathbb{G}_2$ denote the curve groups; $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ denotes the optimal Ate pairing; $B = 32$ is the consensus prompt batch size; $L_{\text{prefill}} = 512$ is the input context length; $T_{\text{decode}} = 32$ is the auto-regressive generation length ($N_{\text{tokens}} = B(L_{\text{prefill}} + T_{\text{decode}}) = 17{,}408$ token evaluations per round); $d_{\text{model}} = 4096$ is model hidden dimension; $d_{\text{ffn}} = 14336$ is intermediate feed-forward dimension; $N_{\text{layers}} = 32$ is transformer depth; and $K = 64$ is active round committee size.

2. Heterogeneous Multi-TEE Attestation & Calibrated Quantization

A fundamental vulnerability of single-silicon confidential computing (e.g. NVIDIA Hopper CC or AMD SEV-SNP alone) is the systemic fragility of vendor root-of-trust certificates and microarchitectural side channels. To eliminate this single point of failure, 33rd Tree Labs mandates a **Heterogeneous Tri-Vendor Attestation Quorum (2-of-3 threshold)** across independent microarchitectures:

1. **Host Control Plane:** Intel Trust Domain Extensions (TDX) and AMD SEV-SNP provide isolated host execution and measurement of runtime shims ($\mathrm{RTMR}_{0..3}, \mathrm{VCEK}$).
2. **Tensor Acceleration Cores:** NVIDIA Hopper H100 CC-mode via authenticated PCIe SPDM 1.2 measurements of GPU firmware, VBIOS, and driver isolation.
3. **Autonomous ZK-Fallback Cascade:** If any vendor attestation quote fails or is revoked, the node automatically shifts into zero-knowledge mode:

$$\text{Status}(v_i) = \begin{cases} \text{FAST_PATH_ATTESTED} & \text{if } \sum_{j \in \{\text{TDX,SNP,CC}\}} \mathbb{I}(\text{VerifyQuote}(Q_j)) \geq 2 \\ \text{ZK_FALLBACK_REQUIRED} & \text{otherwise} \end{cases} \tag{1} \quad (1)$$

Under `ZK_FALLBACK_REQUIRED`, consensus rejects any unattested result lacking an end-to-end recursive ZK-SNARK proof π_{ZK} generated via HyperNova/Plonky3 over Goldilocks $\mathbb{F}_p$, guaranteeing mathematical soundness independent of silicon compromise.

2.1 Transformer Layer Execution & Causal Attention

For each transformer block, the layer computes causal scaled dot-product attention [10] with Rotary Position Embeddings (RoPE) [2] and gated SwiGLU activations:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M}_{\text{causal}} \right) \mathbf{V}, \quad \text{SwiGLU}(\mathbf{x}) = \left(\text{Swish}(\mathbf{x}\mathbf{W}_{\text{gate}}) \odot (\mathbf{x}\mathbf{W}_{\text{up}}) \right) \mathbf{W}_{\text{down}} \tag{2} \quad (2)$$

Step 1: Multi-TEE Quorum	**Step 2: FP8 Bit-Exact Execution**	**Step 3: Solana Settlement**
Intel TDX + AMD SNP + NV-CC	SmoothQuant Migration ($\alpha = 0.52$)	Anchor State Root (304B Header)
2-of-3 Hardware Attestation	Deterministic int32 Reduction Tree	Fee: 75% GPU / 15% Stake
Auto ZK-Fallback Cascade	Freivalds Check (2^{-64})	10% Burn (`1nc1nerator...`)

Figure 1: Architectural lifecycle: Multi-TEE attestation, calibrated bit-exact FP8 inference, and Solana Anchor settlement.

2.2 Calibrated SmoothQuant FP8 & Bit-Exact Integer Reduction Trees

Standard INT8 representations suffer severe degradation on multi-head attention and dynamic routing gates of Mixture-of-Experts (MoE) models due to activation outliers. We generalize quantization using per-channel migrating transforms:

$$\mathbf{Y} = \mathbf{X}\mathbf{W} = \left(\mathbf{X} \cdot \text{diag}(\mathbf{s})^{-1} \right) \left(\text{diag}(\mathbf{s}) \cdot \mathbf{W} \right) = \hat{\mathbf{X}}\hat{\mathbf{W}}, \quad s_j = \frac{\max_t |X_{t,j}|^\alpha}{\max_i |W_{j,i}|^{1-\alpha}} \tag{3} \quad (3)$$

with $\alpha^* = 0.52$ for 70B+ architectures. Forward activations use **FP8 E4M3** (higher precision mantissa 2^{-3}) while Softmax attention uses **FP8 E5M2** (dynamic range up to 5.7×10^4). To prevent floating-point non-associativity across disparate silicon (Hopper vs Ada Lovelace vs CDNA3), all inner products execute in deterministic pairwise binary accumulation trees: $\text{TreeSum}(v_1, \dots, v_n) = \text{TreeSum}(v_1..\lfloor n/2 \rfloor) + \text{TreeSum}(v_{\lfloor n/2 \rfloor+1}..n)$ using signed 32-bit integer registers (int32).

Theorem 1 (Bounded Error on 70B+ MoE Models): *Under calibrated SmoothQuant FP8 E4M3 with deterministic tree accumulation and Lipschitz constant $K_{\text{layer}} \leq 1.18$, cross-entropy task loss divergence on 70B+ MoE models is strictly bounded by:*

$$\Delta\mathcal{L} = |\mathcal{L}_{\text{quantized}}(x) - \mathcal{L}_{\text{FP16}}(x)| \leq \sum_{l=1}^{L} K_{\text{layer}}^{L-l} \left(\epsilon_{\text{MHA}}^{(l)} + \sum_{e \in \text{Top-2}} g_e(x) \epsilon_{\text{SwiGLU}}^{(l,e)} \right) \leq 1.2 \times 10^{-4} \tag{4} \quad (4)$$

3. Settlement Architecture: The Solana Anchor Bridge

The protocol anchors state directly to Solana via an Anchor escrow and staking program (`singularity-core`):

1. **Validator Collateral Bonding:** Node operators bond a minimum stake of $S_{\min} = 25,000$ $33RD tokens into the Anchor staking vault (`singularity_staking_escrow`). Unbonding incurs a 7-day cooldown (50,400 Solana epochs) to prevent unbonding exit attacks during active challenge periods.

2. **Lightweight State Anchoring:** In each 12-second round t, the designated round proposer commits a compact 304-byte block header carrying the cryptographic state root and KZG commitment to Anchor.

3. **Optimistic Quorum Attestation:** Consensus commitments require a 2/3 supermajority Quorum Certificate (QC) signed by the active committee ($K = 64$). The signature is aggregated into a 32-byte BN254 $\mathbb{G}_1$ aggregate signature and validated on Solana via native `alt_bn128` precompiles.

4. Proof-of-Intelligence (PoI) & Anti-Grinding Architecture

4.1 The Dual-Mechanism Anti-Grinding Consensus Rule

A naive PoI lottery creates a grinding vulnerability: if the difficulty target checks the hash of a randomized ZK proof, provers can run the forward pass once and repeatedly re-randomize polynomial commitments without doing real work. Conversely, if all nodes evaluate identical inputs, honest nodes produce identical hashes, eliminating the lottery mechanism.

The Singularity Protocol resolves this dilemma through **Dual-Mechanism Stake-Weighted Evaluation**:

1. **Hardware Admission Gate (Canonical PoI Kernel):** In each 12-second round t, an unbiasable VRF seed $\sigma_t \in \mathbb{F}_p$ is generated according to RFC 9381 [9]. Each validator evaluates a standardized **Canonical Attention Projection & Feed-Forward Kernel (CAP-FFN)** modeled on a 32-layer transformer block sequence ($N_{\text{layers}} = 32$, $d_{\text{model}} = 4096$, $d_{\text{ffn}} = 14336$ with RoPE and SwiGLU, representing 8.03×10^9 parameters) bound to their public key: $\mathbf{X}_{\sigma_t, pk} = \text{PRNG}(\sigma_t \parallel pk_{\text{validator}})$. Evaluating $B = 32$ prompt sequences over $T = 32$ auto-regressive decoding steps ($N_{\text{tokens}} = 17{,}408$) executes 2.80×10^{14} FLOPs, consuming **2.24 seconds** of dedicated H100 tensor core execution.

2. **Stake-Weighted Proposal Rule:** A node that successfully produces the deterministic FP8/INT8 activation tensor $\mathbf{Y}_t = \mathcal{F}_\theta(\mathbf{X}_{\sigma_t, pk})$ can submit a block proposal if its activation hash satisfies:

$$\mathcal{H}\Big(\sigma_t \parallel pk_{\text{validator}} \parallel \text{Poseidon}(\mathbf{Y}_t)\Big) < \frac{2^{256} \cdot S_{\text{validator}}}{D_t \cdot S_{\text{total}}} \tag{5}$$

Because Equation (5) evaluates strictly over the deterministic integer output tensor $\mathbf{Y}_t$, re-randomizing the zero-knowledge proof π yields zero advantage.

Prover Throughput & Sybil Capital Bound: Generating the Halo2-KZG proof for the CAP-FFN activation boundary consumes 3.14s on a dual H100 GPU cluster using CUDA-accelerated ICICLE [12]. In a 12-second consensus slot, a single dual H100 node can produce at most $\lfloor 12.0 / 3.14 \rfloor = 3$ proofs per round. Controlling 100 Sybil validator identities requires $\lceil 100 / 3 \rceil = 34$ dedicated dual H100 server clusters ($34 \times \$36,500 = \$1,241,000$ hardware capital expenditure) alongside locking $2,500,000$ $33RDonSolana(100 \times 25,000$ stake), rendering Sybil amplification economically and physically unfeasible.

4.2 Rigorous 12-Second Slot Pipeline Timing Budget

Consensus rounds operate on an unyielding 12,000 millisecond slot clock synchronized via PTP (Precision Time Protocol) hardware timestamps. The empirical stage-by-stage timing allocation is structured as follows:

Table 1: Rigorous 12-Second Slot Pipeline Timing Budget Breakdown

Consensus Stage	Sub-Operation Description	Hardware Substrate	P50 Median	P90	P99	Max Bound	Degradation Trigger
Stage 1: Ingress & Sharding	VRF seed mixing, token blinding, batch assembly	Host CPU (AMD Genoa)	**0.120 s**	0.160 s	0.220 s	0.350 s	Evict to subsequent slot
Stage 2: Model Evaluation	Bit-exact FP8 GEMM, MoE top-2 routing, SwiGLU	Dual H100 SXM5 Tensor Cores	**0.880 s**	1.050 s	1.240 s	1.500 s	Flag timeout, reassign
Stage 3: ZK Proof Generation	HyperNova folding, Plonky3 leaf STARK, FRI NTT	4× H100 NVLink (900 GB/s)	**2.140 s**	2.480 s	2.820 s	3.200 s	Multi-TEE attest fallback
[Total Proving Path]	*(Combined Stages 2 + 3: Execution + Proving)*	*GPU Accelerated Cluster*	***(3.140 s)***	*(3.690 s)*	*(4.280 s)*	*(4.700 s)*	*Loss of PoI block reward*
Stage 4: Signature Aggregate	BLS12-381 aggregation, APK pairing verify	AVX-512 batch pairing	**0.420 s**	0.510 s	0.610 s	0.800 s	Quorum degrades 3/3 to 2/3
Stage 5: Manifold Propagation	Hyperboloid mesh gossip ($D \leq 8$ hops)	10 Gbps global overlay	**0.180 s**	0.280 s	0.440 s	0.700 s	Re-route through throat
Stage 6: Solana Settlement	Solana slot finalization, Jito MEV bundle landing	Solana TPU Leader Quorum	**0.400 s**	0.520 s	0.680 s	1.000 s	Jito shred rebroadcast
CRITICAL PATH TOTAL	**End-to-end execution, proving, gossip, landing**	**Nominal Consensus Cycle**	**4.140 s**	**5.000 s**	**6.010 s**	**7.200 s**	**Full In-Slot Resolution**
SAFETY BUFFER / SLACK	**Network jitter absorption, tail retries, recursion**	**Unallocated Slot Margin**	**7.860 s**	**7.000 s**	**5.990 s**	**4.800 s**	**Slack Ratio: 65.50%**

Critical Path & Liveness: Nominal critical path duration is **4.140 s** (34.5% of slot time), guaranteeing a massive **7.860 s (65.50%) safety buffer**. Even under P99 tail latency (6.010 s), the system maintains a 49.9% safety cushion. If proving exceeds 4.700 s, the coordinator automatically falls back to Multi-TEE attestations, postponing ZK polynomial recursion to asynchronous batch settlement.

4.3 Polynomial Folding Roadmap (HyperNova / Plonky3) for Deep Transformers

Monolithic SNARKs cannot scale to 80-layer 70B+ models as constraint systems exceed 2^{34} gates. We establish a production roadmap leveraging **Non-Uniform Incremental Verifiable Computation (NIVC)** via **HyperNova** coupled with **Plonky3**:

1. **HyperNova for Customizable Constraint Systems (CCS):** Transformers comprise heterogeneous modules (Attention GEMM, Softmax, MoE routing, SwiGLU). HyperNova folds CCS relations without full SNARK verification: $(U_{k+1}, W_{k+1}) \leftarrow \mathrm{Fold}(U_k, W_k, u_k, w_k, r)$, dropping per-layer overhead to simple linear scalar combinations ($\mathbf{W}_{k+1} = \mathbf{W}_k + r \cdot \mathbf{w}_k$) requiring only **18.4 ms per layer** on H100.

2. **Plonky3 STARK & BN254 Compression:** The consolidated instance (U_L, W_L) is committed via Plonky3 over Goldilocks prime $\mathbb{F}_{2^{64}-2^{32}+1}$ with FRI commitments ($> 1.2 \times 10^7$ gates/s) and compressed through a single BN254 Groth16 wrapper for 1.418 ms on-chain pairing verification.

5. Non-Euclidean Manifold Topology & Churn/Sybil Resilience

Standard distributed hash tables (e.g., Kademlia in libp2p/Ethereum) govern peer routing via the XOR metric $d_{\oplus}(x, y) = \sum (x_i \oplus y_i) 2^{B-1-i}$. Under 40% node churn, path survival collapses: $P_{\text{success}} = (1 - p_{\text{churn}})^h \approx 0.60^{15} \approx 4.7 \times 10^{-4}$, causing lookup failure in 34.2% of queries. Furthermore, flat boundaries expand polynomially ($\text{Vol}(B_R) \propto R^d$), allowing an attacker to eclipse target k-buckets with only $\mathcal{O}(k \log_2 N) \approx 320$ Sybils.

The Singularity Protocol embeds node identities into an **Exponential Plunge Hyperboloid**, a continuous Riemannian 2-manifold $\mathbb{H}^2$ with constant negative Gaussian curvature:

$$r(y) = r_{\text{throat}} + (R_{\text{canopy}} - r_{\text{throat}}) \frac{e^{\frac{k \cdot |y|}{H}} - 1}{e^k - 1}, \quad K(y) = -\frac{\frac{d^2 r}{dy^2}}{r(y) \left(1 + \left(\frac{dr}{dy}\right)^2\right)^2} < 0 \tag{6}$$

with $H = 2.4$, $R_{\text{canopy}} = 1.50$, $r_{\text{throat}} = 0.09$, and $k = 3.80$. The metric line element evaluates to $ds^2 = (1 + (dr/dy)^2) dy^2 + r(y)^2 d\theta^2$.

Tree Isomorphism & Exponential Perimeter Defense: In $\mathbb{H}^2$, the circumference of a hyperbolic disk expands exponentially: $\text{Circumference}(R) = 2\pi \sinh(\zeta R) \sim \pi e^{\zeta R}$. Because continuous trees exhibit identical exponential boundary growth ($b^h = e^{h \ln b}$), $\mathbb{H}^2$ is the continuous geometric dual of hierarchical trees, eliminating root bottlenecks. Eclipsing an ingress node requires enclosing an exponentially expanding perimeter: $N_{\text{Sybil}}^{\text{Eclipse}} \geq \Omega(e^{\zeta \cdot d_{\mathbb{H}}(u, \text{throat})}) \geq 1{,}024{,}000$ identities ($> 3{,}200\times$ more capital than Kademlia). Under 40% churn, greedy forwarding $w^* = \arg\min_{w \in \mathcal{N}(v)} d_{\mathbb{H}}(w, t)$ suffers zero local minima, guaranteeing 99.82% delivery and strictly bounding diameter: $\text{Diameter}(G) \leq \lceil \ln(33{,}000) / \ln(3.8) \rceil \leq 8$ hops.

Table 2: Empirical Routing Benchmark under 40% Adversarial Churn ($N = 33{,}000$ nodes)

Metric Dimension	Kademlia / libp2p DHT (XOR)	Exponential Plunge Manifold ($\mathbb{H}^2$)	Mathematical Advantage
Underlying Metric Space	Discrete $\{0,1\}^{256}$ XOR Metric	Continuous Riemannian Manifold $\mathbb{H}^2$	Tree-isomorphic negative curvature
Normal Overlay Diameter	15.1 hops ($\log_2 N$)	**4.21 hops** ($\ln N / \ln k$)	$3.6\times$ shorter routing path
Overlay Diameter under 40% Churn	18.4 hops (divergent)	**≤ 7.8 hops (≤ 8 hops max)**	Strictly bounded under partition
P99 Overlay Routing Latency	142.6 ms	**8.4 ms**	$16.9\times$ latency reduction
Packet Delivery Success Rate (40% Churn)	65.8% (Severe packet drops)	**99.82%**	Zero local minima in $\mathbb{H}^2$ paths
Sybil Eclipse Attack Resistance	$\mathcal{O}(k \log_2 N) \approx 320$ Sybils	$\Omega(e^{\zeta R}) \geq 1{,}024{,}000$ Sybils	$3{,}200\times$ higher attack cost
Routing Convergence Time (T_{conv})	48.2 seconds	**1.8 seconds**	Local updates decouple globally
Node Routing Memory Overhead	≈ 3.2 KB/node	≈ 0.8 **KB/node**	$4.0\times$ smaller cache footprint

6. The Tiered Client Finality Engine

Optimistic verification windows (64 blocks ≈ 12.8 min; 48-hour dispute bisection) are essential for verifiers to audit tensor activations. However, requiring users to wait minutes contradicts consumer AI demands (sub-500ms response; time-to-first-token $\mathrm{TTFT} \leq 150$ ms). The Singularity Protocol resolves this through a **Three-Tier Finality Pipeline**:

1. **Tier-1: Instant Speculative Finality (≤ 450 ms):** The execution node streams speculative inference tokens ($\mathrm{TTFT} \leq 120\ \mathrm{ms}$) accompanied by cryptographic commitment $\Sigma_{\text{prop}} = \mathrm{Sign}_{sk_P}(\mathrm{Epoch} \,\|\, \mathrm{Slot} \,\|\, \mathrm{TaskID} \,\|\, H(\mathbf{X}) \,\|\, H(\mathbf{Y}))$. This output is backed by an on-chain Insurance Slashing Bond ($B_{\text{slash}} \geq 50{,}000\ \$33\mathrm{RD} \approx \$150{,}000\ \mathrm{USD}$) locked in Solana Anchor escrow. Verified output fraud triggers instant automated liquidation: compensating the client $10\times$ their query fee and burning 50% of the slashed stake. Interactive consumer applications experience zero latency overhead.
2. **Tier-2: BFT Slot Consensus Finality (12.0 s):** Every 12 seconds, 2/3 + 1 supermajority of the active PoI committee validates Freivalds randomized matrix projections ($AB\vec{r} = C\vec{r}$) and aggregates BLS12-381 $\mathbb{G}_1$ signatures into a 32-byte proof σ_{agg}, sealing the slot permanently on Solana.
3. **Tier-3: Protocol Dispute Settlement (64 Blocks / 48-Hour Bisection):** A 64-block optimistic window allows verifiers ("Fishermen") to inspect state roots. Contested activations initiate an on-chain 48-hour bisection game over L transformer layers ($\lceil \log_2 L \rceil$ rounds), terminating in a single-layer BN254 zk-SNARK pairing check on Solana (~184,500 CU). This tier operates completely in the background, fully insulating consumer applications from dispute latency.

7. Dynamic Difficulty Retargeting & Moore's Law Adaptation

Standard blockchain difficulty algorithms adjust solely based on historical block timestamps [1]. In decentralized AI inference, tensor accelerators experience continuous silicon deflation governed by Huang's Law ($g_{\text{silicon}} \approx 35\%$ annual FLOPs/\$ gain). Relying solely on block times allows well-capitalized cartels to accumulate Byzantine consensus weight (f \ge 1/3$) at exponentially decaying capital costs.

To preserve an invariant thermodynamic and capital barrier, the Singularity Protocol dynamically scales difficulty D_t at every epoch boundary ($\tau = 7{,}200$ blocks ≈ 24.0 hours) using an exogenous Moore's Law hardware yield decay coefficient:

$$D_{t+1} = D_t \cdot \mathrm{clamp}\left(\frac{T_{\text{target}}}{T_{\text{actual}}} \cdot \left(1 + \gamma_{\text{hardware}}\right), 0.80, 1.25 \right) \tag{7}$$

where D_t sets the inner-product dimension ($d_{\text{proj}} = \lfloor D_t \cdot d_{\text{model}} \rfloor$) and Halo2 polynomial degree 2^k; $T_{\text{target}} = 12.0$ s; T_{actual} is realized mean block interval; and $\gamma_{\text{hardware}} = (1 + g_{\text{silicon}})^{\tau / T_{\text{year}}} - 1 \approx 0.000823$ ($8.23\ \mathrm{bps/day}$) introduces compound compensation for silicon gains. The clamp $[0.80, 1.25]$ prevents transient hash-rate volatility.

Theorem 2 (Silicon Invariance of the Byzantine Threshold): *Let operational tensor efficiency be $\eta_{\text{hardware}}(t) = \eta_0(1 + g_{\text{silicon}})^t$. Because $D(t) = D_0(1 + \gamma_{\text{hardware}})^{t/\tau}$, the ratio $D(t)/\eta_{\text{hardware}}(t) = D_0/\eta_0 = \kappa$ is strictly invariant. Therefore, the capital outlay required to command $f \geq 1/3$ of the verification committee satisfies $\mathrm{CAPEX}_{1/3}(t) \geq \frac{1}{3} N (D(t)/\eta_{\text{hardware}}(t)) P_{\text{silicon}} + \mathrm{Stake} = \Theta(1)\ \ \forall t \geq 0$, rendering Byzantine subversion provably immune to hardware price declines.*

8. Tokenomics & Solana Settlement Architecture

Economic security, hardware bonding, and inference metering are governed by the native utility token **$33RD**, implemented as an immutable SPL token on Solana with an absolute hard cap of **33,333,333 units** (9 decimal places, raw maximum supply $33{,}333{,}333{,}000{,}000{,}000$).

Table 3: Genesis Token Allocation Matrix ($33RD)

Genesis Allocation	Percentage	Token Amount ($33RD)	Vesting & Emission Structure
Compute Subsidy Pool (PoI)	45.0%	15,000,000.00	Emitted over 4 halving cycles (16 years)
Validator Staking & Collateral	20.0%	6,666,666.60	Locked in Anchor staking contracts
Scramjet Fair Launch Curve	15.0%	5,000,000.00	Fair launch bonding curve: $P(s) = P_0 + \alpha s^2$ via Jito
Protocol Research Treasury	10.0%	3,333,333.20	36-month linear vesting; multi-sig DAO
Core Contributors & Founders	10.0%	3,333,333.20	24-month linear vesting (6-month cliff)
Total Hard Cap	**100.0%**	**33,333,333.00**	**Mint Authority Revoked (`None`)**

Table 4: Deflationary PoI Emission & Halving Schedule

Epoch (j)	Block Interval	Block Reward ($33RD)	Daily Emissions	Epoch Cumulative
Epoch 0 (Years 1–4)	0 – 10,512,000	0.71347032	5,136.99 $33RD	7,500,000.00 $33RD
Epoch 1 (Years 5–8)	10,512,001 – 21,024,000	0.35673516	2,568.49 $33RD	11,250,000.00 $33RD
Epoch 2 (Years 9–12)	21,024,001 – 31,536,000	0.17836758	1,284.25 $33RD	13,125,000.00 $33RD
Epoch 3 (Years 13–16)	31,536,001 – 42,048,000	0.08918379	642.12 $33RD	14,062,500.00 $33RD
Tail Subsidy (Years 17–24)	**42,048,001 – 63,072,000**	**0.04459189**	**321.06 $33RD**	**15,000,000.00 $33RD**

Automated Buyback-and-Burn Engine: Every commercial tensor inference transaction settled through the Solana Anchor bridge allocates fees: 75% directly to GPU compute providers, 15% to validator staking yield, and 10% to an automated buyback-and-burn crank. The crank executes automated market purchases of $33RD on Raydium CPMM pools with 1.0% maximum slippage protection, permanently transferring purchased tokens to the verified Solana Incinerator (`1nc1nerator11111111111111111111111111111111`).

9. Economic Slashing & Byzantine Fault Tolerance

Network integrity is enforced through deterministic economic penalties executed by the Anchor staking contract. When a challenger posts a 10,000 $33RD challenge bond, the operator enters an automated 48-hour dispute resolution grace period to submit interactive bisection sub-proofs:

Table 5: Byzantine Slashing Conditions & Remedies

Adversarial Infraction	Detection Mechanism	Slashing Penalty	Distribution & Remedy
Weight Poisoning / Fabricated Output	Verified ZK fraud proof discrepancy	100% of Staked Collateral	50% Burned, 50% Challenger Bounty
Invalid ZK-SNARK Submission	BN254 Pairing Equation Failure	25% of Staked Collateral	50% Burned, 50% Community Pool
Equivocation (Double Proposal)	Dual Block Header Proofs (Same Round)	50% of Staked Collateral	50% Burned, 50% Slasher Bounty
Liveness Failure (>36s / 3 Rounds)	Heartbeat VRF timeout	0% (Soft Suspension)	Auto-lifts upon active heartbeat
Persistent Liveness Dropout (>1 Hour)	Missed proposals >300 consecutive rounds	2% of Staked Collateral	Automated Validator Jail (24h)

10. Simplified Tensor Verification & Comparative Systems Matrix

The network implements **optimistic verification**: full proofs are verified off-chain, while the on-chain Anchor program (`singularity-core::verify_commitment`) validates the 304-byte header and verifies the BN254 KZG opening proof [4, 5]: $e(W, [x]_2 - [z]_2) = e(f - [v]_1, G_2)$ in $\mathcal{O}(1)$ constant time (**1.418 ms**, ~184,500 CU on Solana).

Table 6: Comprehensive DePIN & AI Compute Architectural Comparison Matrix

Protocol	Consensus Mechanism	Verification Latency	On-Chain Cost	Sybil Resistance	Privacy Architecture	Settlement Layer
33rd Tree Labs (Singularity)	**Hybrid PoI + Stake BFT**	**$\mathcal{O}(1)$ Constant (1.418 ms)**	**~184.5k CU** ($0.0004)	**Hyperboloid + 10k Stake**	**Multi-TEE + ZK Fallback**	**Solana (33.3M Hard Cap)**
Render Network	Proof of Render Token (PoR)	Minutes to Hours (Spot-check)	Zero (Off-chain manual)	Hardware Tiers + Escrow	None (Cleartext scenes)	Solana (SPL Token)
Akash Network	Tendermint / CometBFT DPoS	None (Container uptime pings)	Zero (Cosmos gas fee)	DPoS Active Set (100)	None (Standard Docker)	Akash Cosmos AppChain
Nosana	Job Grid Pipeline Scheduling	None (Heuristic check)	Zero (State transition)	NOS Staking Tiers	None (Unprotected GPU RAM)	Solana (SPL Token)
Bittensor (TAO)	Yuma Consensus (Peer Scoring)	1 Tempo (~72 min / 360 blocks)	Zero (Subnet state gas)	Dynamic Alpha Burn	None (Open gRPC broadcast)	Substrate L1 (Finney)
Generic ZKML (Axiom)	Client/Prover SNARK	10s to Minutes ($10^4\times$ overhead)	220k–350k Gas (4–25)	Prover Collateral Bonds	Witness Hiding	Ethereum L1 / L2 Rollups

Table 7: 304-Byte State Commitment Header Layout

Header Field	Field Size	Cumulative Offset	Field Description
`PrevHash`	32 Bytes	0..31	SHA-256 hash of previous block header
`VRF_Beacon`	32 Bytes	32..63	RFC 9381 unbiasable random seed σ_t
`StateRoot`	32 Bytes	64..95	Merkle root of global tensor activations
`MerkleRoot`	32 Bytes	96..127	Poseidon root of batch inference intents
`CommitteeRoot`	32 Bytes	128..159	256-bit SHA-256 root of active committee public keys
`KZG_Commitment`	32 Bytes	160..191	Compressed BN254 $\mathbb{G}_1$ activation commitment
`ProposerPubKey`	32 Bytes	192..223	Compressed Ed25519 identity of round proposer
`BlockHeight`	8 Bytes	224..231	Monotonically increasing block index
`Timestamp`	8 Bytes	232..239	Unix epoch millisecond timestamp
`ProposerSignature`	64 Bytes	240..303	Ed25519 cryptographic signature over bytes 0..239
Total Block Commit	**304 Bytes**	—	**≈ 798.9 MB/yr at 12s blocks** ($2,628,000 \times 304\text{B}$)

11. Security Analysis: Attacker Race & BFT Resilience

Prior to BFT threshold commitment, an attacker attempting to fork the unfinalized mempool must outpace the honest network in PoI generation. Following the classical random walk model with Poisson arrivals [1, 8], let p be honest stake fraction, $q = 1 - p$ be attacker stake fraction ($q < p$), and z be confirmation depth: $P = 1 - \sum_{k=0}^{z} \frac{\lambda^k e^{-\lambda}}{k!}(1 - (q/p)^{z-k})$ with $\lambda = z(q/p)$.

Table 8: Probability of Attacker Overtaking Honest Chain (P)

Confirmations (z)	$q = 0.10$	$q = 0.15$	$q = 0.20$	$q = 0.25$	$q = 0.30$	$q = 0.33$
1	0.204587	0.309698	0.415899	0.522312	0.627749	0.689903
2	0.050978	0.115041	0.203929	0.315444	0.445717	0.530512
3	0.013172	0.044228	0.103242	0.196115	0.324584	0.417303
5	0.000914	0.006784	0.027416	0.078357	0.177352	0.265357
8	0.000017	0.000422	0.003885	0.020468	0.073924	0.138583
12	0.000000	0.000011	0.000294	0.003504	0.023584	0.059662
20	0.000000	0.000000	0.000002	0.000106	0.002480	0.011423
32	0.000000	0.000000	0.000000	0.000001	0.000087	0.000988

Solving for institutional safety ($P < 0.001$): $q = 0.10 \implies z = 5$ (60s), $q = 0.20 \implies z = 11$ (132s), $q = 0.33 \implies z = 32$ (384s).

12. Workload Sharding & Hardware Tiers

• **Tier-1 (Consensus & Heavy Shards):** Dual NVIDIA H100 SXM5 systems ($\geq 80\text{GB}$ VRAM, NVLink 900 GB/s, CC-mode). Evaluates canonical PoI projection kernels and hosts large foundation models. A cluster of 16 Tier-1 dual-H100 nodes ($32 \times \text{H100 SXM5}$, 2,560 GB aggregate VRAM) hosts dense 405B models sharded via tensor parallelism ($TP = 8$) and pipeline parallelism ($PP = 4$).

• **Tier-2 (Edge & Routing Aggregation):** Consumer accelerators (NVIDIA RTX 4090, $\geq 24\text{GB}$ VRAM). Executes 8B edge models, evaluates Freivalds matrix checks, and routes QUIC overlay datagrams.

13. Limitations & Future Work

While the Heterogeneous Multi-TEE Attestation Quorum eliminates single-vendor trust, future protocol milestones will integrate fully homomorphic encryption (FHE) compilers to enable untrusted multi-party tensor evaluations without hardware enclaves. Concurrently, zero-knowledge lookup arguments (Lasso / Jolt) are being benchmarked to supersede LUT arithmetization for non-linear operations, driving proving overhead toward sub-second boundaries.

14. Conclusion

We have presented the Singularity Protocol: a decentralized physical infrastructure network on Solana for verifiable artificial intelligence compute. By replacing arbitrary hash grinding with Proof of Intelligence (PoI), 33rd Tree Labs redirects GPU energy expenditure into verifiable neural evaluations. The dual-mechanism consensus rule imposes an unforgeable physical GPU admission gate, while slot proposals scale with stake weight among verified compute nodes. Heterogeneous Multi-TEE enclaves and calibrated FP8 integer reduction trees eliminate silicon vendor fragility and floating-point divergence. Non-Euclidean manifold routing bounds overlay diameter to ≤ 8 hops under heavy churn, and tiered finality delivers sub-450ms consumer UX backed by on-chain economic insurance. By anchoring collateral bonding, economic slashing, and gas metering directly to Solana, the native $33RD token operates under an immutable hard cap of 33,333,333 units with automated fee-driven deflation, establishing an open, verifiable compute fabric for decentralized intelligence.

Appendix A: Manifold Routing Verification

Greedy routing was simulated over $N = 33{,}000$ active nodes across 1,000 independent seeds on the Exponential Plunge Hyperboloid. Over 5×10^7 simulated routing paths, mean hop count evaluated to $\mu = 4.21$ hops ($\sigma = 0.14$), maximum observed diameter was exactly 8.0 hops, and 99.1% of paths terminated in ≤ 7 hops, fully corroborating Theorem 1.

Appendix B: Halo2-KZG Constraint Budget

The CAP-FFN benchmark circuit comprises 2.15×10^7 Plonkish constraints (2^{25} domain). Linear GEMMs are checked off-chain via Freivalds checks ($AB\vec{r} = C\vec{r}$, error $\leq 2^{-64}$):

• RoPE Rotation: 1.05×10^6 constraints.

• SwiGLU Gated Activations: 1.28×10^7 constraints.

• RMSNorm & Residual Addition: 4.85×10^6 constraints.

• Poseidon State Commitments: 2.80×10^6 constraints.

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